Basem Saleh

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Summary

Senior Machine Learning Engineer with 5+ years at Lockheed Martin spanning computer vision, reinforcement learning, NLP, and production MLOps. Holds an M.S. in Artificial Intelligence from Johns Hopkins University. Proven track record deploying ML systems from research to internal products; currently expanding expertise in agentic LLM systems and multi-agent orchestration.

Professional Experience

Lockheed Martin — Moorestown, NJ
Senior Machine Learning Engineer — May 2024 - Present

- Trained a multi-agent MLP-based, reinforcement learning system in a JAX environment, achieving 20% faster mission completion vs. a deterministic baseline through learned cooperative behaviors under scenario constraints.
- Contributed to a RAG-powered LLM assistant using Weaviate that generated Cameo architectures compliant with design standards, automatically placing outputs into engineers' workspaces.
- Designed a constraint satisfaction algorithm for an autonomous system to evaluate mission feasibility against environment variables and operational constraints, enabling reliable go/no-go mission decisions.
- Led a small team to automate simulations and data collection via KubeFlow, reducing a 1-day manual process to an overnight, automatic pipeline with LakeFS transfers — enabling parallel runs without engineer oversight.

Lockheed Martin — Moorestown, NJ
AI Engineer — November 2023 - May 2024

- Classified key customer frequencies with 98% accuracy using real-time, digital signal processing machine learning Pytorch models, contributing to securing new funding for the project's second phase.
- Spearheaded a regression model to predict Jira point estimations from ticket metadata and team labels, reducing program schedule overruns — later adopted and iterated into an official, internal product.
- Authored an internal white paper evaluating efficacy of object detectors over image classification architectures in wideband signal applications.
- Enabled real-time field response capabilities by developing a supervised classification model that identified 5–6 wideband UAV signal types with 95% accuracy using PyTorch.

Lockheed Martin — Moorestown, NJ
Software Engineer — July 2019 - November 2023

- Slashed radio hardware validation time from 1 hour to minutes by building a GitLab CI/CD pipeline that automated containerized builds, Valgrind memory checks, and data testing for a 4-person team.
- Saved over $70,000 in radar project costs by refactoring legacy architecture of a radar error model.
- Reduced radar panel calibration time from one day to 10 minutes using C# to control an oscilloscope and switch matrix for S-parameter measurements.

Research Papers and Projects

Paper: Keyboard Keystroke Predictions on Various Switches — May 2024 - Aug 2024
Developed and trained a custom CoAtNet fusion model to utilize Computer Vision to predict keyboard keystrokes from keyboard audio and features to achieve over 96% accuracy for single-key classification.

Project: Game Reinforcement Learning AI Agent with Object Detection — Dec 2024 - Present
Combining YOLOv8 object detection models, software-driven keyboard and mouse inputs, and reinforcement learning to train an AI agent to play Dead By Daylight, a 3D multiplayer game.

Education

Master of Science in Artificial Intelligence, Johns Hopkins University — Dec 2024
Bachelor of Science in Electrical Engineering, Drexel University — June 2020

Skills

Programming Languages:	Python
Frameworks & Libraries:	PyTorch, JAX, Hugging Face Transformers, OpenCV, Scikit-Learn, Streamlit, TensorBoard
AI & Machine Learning:	Reinforcement Learning, Computer Vision, Retrieval-Augmented Generation (RAG), LLM Applications, Embedding Models, Vector Databases
MLOps & Infrastructure:	Kubeflow, MLflow, LakeFS, Weights & Biases (W&B), Docker, GitLab CI/CD, AWS, Linux
Spoken Languages:	English, Arabic (Egyptian)